Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD Chemist with 6+ years of analytical and materials characterization experience across organic materials, polymers, thin films, and stability-sensitive process-development programs. Brings LC-MS/NMR/FT-IR/UV-Vis/GPC/thermal analysis, DOE/SPC, SOP-style lab operations, and technical reporting discipline to pharmaceutical analytical development teams.

CORE COMPETENCIES

Analytical Characterization | Method Development & Troubleshooting | LC-MS / Chromatography-Adjacent Workflows | NMR / FT-IR / UV-Vis / GPC | Thermal Analysis (TGA/DSC) | Stability & Failure-Mode Analysis | DOE / SPC | Technical Reports & Presentations

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Leading DOE-guided process development for bio-based polymer systems, defining critical parameters, improving reproducibility, and establishing scale-up-ready process specifications.
- Building characterization workflows, SOP-style records, and technical reports that connect analytical data with stability-sensitive process and product-development decisions.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Built multi-technique analytical workflows using FT-IR, UV-Vis-NIR, cyclic voltammetry, NMR, TGA, and DSC to characterize 15+ organic and organometallic systems and interpret performance-limiting mechanisms.
- Troubleshot analytical workflows, interpreted complex datasets, and translated results into manuscripts, presentations, and decision-ready technical conclusions.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded thin-film process development, building reproducible deposition and characterization workflows for high-performance molecular coatings in a multidisciplinary setting.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Developed CVD-based polymer thin-film deposition processes using DOE and in-situ QCM-D monitoring to improve process control and reproducibility.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed laboratory operations for a 15-member research group, improved instrument uptime by 30%, and wrote SOP-style documentation to support reliable experimental execution.

Georgia Institute of Technology

2019 - 2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates and developed supplemental materials that improved exam performance by 15%, strengthening technical training and communication skills.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Bio-Based Polymer Scale-Up Process

Current

Running DOE-guided process optimization and characterization studies to define reproducible operating windows and quality-relevant specifications for polymer systems moving toward scale-up.

DOE, process development, characterization, technical documentation

Electrochromic Material Reliability Study

Chemistry of Materials 2025

Built stability and failure-analysis workflows for electrochromic materials, using multi-technique characterization to connect degradation behavior with design and processing choices.

FT-IR, UV-Vis-NIR, cyclic voltammetry, TGA, DSC, root-cause analysis

AFOSR Defense Thin-Film Coatings

DoD Funded

Supported thin-film process development and reproducible characterization for defense-oriented coatings programs in a cross-functional environment with demanding technical timelines.

Thin films, process development, characterization, cross-functional execution

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Analytical Chemistry: LC-MS/MS, NMR, FT-IR, UV-Vis-NIR, GC-MS, GPC/SEC, TGA/DSC, QCM-D, ellipsometry, spectroelectrochemistry
Analytical Development Support: Method development mindset, troubleshooting, stability-oriented testing, technical documentation, SOP-minded lab operations, data interpretation
Statistical Analysis: DOE, Statistical Process Control (SPC, Coursera 2026), JMP (basic), Python (basic), Excel (advanced), data visualization
Materials Expertise: Polymers, thin films, organic and organometallic materials, structure-property-processing relationships, degradation analysis
Project Leadership: Cross-functional execution, multi-project prioritization, stakeholder communication, laboratory coordination, technical presentations

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.

2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)